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Java SE 27 & JDK 27

Interface AnnotatedType

All Superinterfaces:

AnnotatedElement

All Known Subinterfaces:

AnnotatedArrayType, AnnotatedParameterizedType, AnnotatedTypeVariable, AnnotatedWildcardType

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```
public interface AnnotatedType
extends AnnotatedElement
```

AnnotatedType represents the potentially annotated use of a type in the program currently running in this VM. The use may be of any type in the Java programming language, including an array type, a parameterized type, a type variable, or a wildcard type. Note that any annotations returned by methods on this interface are *type annotations* (JLS §9.7.4) as the entity being potentially annotated is a type.

See *Java Language Specification*:

- 4.1 The Kinds of Types and Values
- 4.2 Primitive Types and Values
- 4.3 Reference Types and Values
- 4.4 Type Variables
- 4.5 Parameterized Types
- 4.8 Raw Types
- 4.9 Intersection Types
- 10.1 Array Types

Since:

1.8

Method Summary

All Methods	Instance Methods	
Modifier and Type	**Method**	**Description**
default AnnotatedType	**getAnnotatedOwnerType**()	Returns the potentially annotated type that this type is a member of, if this type represents a nested type.
<T extends Annotation> T	**getAnnotation** (Class<T> annotationClas	Returns this element's annotation for the specified type if such an

		annotation is *present*, else null.
`Annotation[]`	**`getAnnotations`**`()`	Returns annotations that are *present* on this element.
`Annotation[]`	**`getDeclaredAnnotations`**`()`	Returns annotations that are *directly present* on this element.
`Type`	**`getType`**`()`	Returns the underlying type that this annotated type represents.

Methods declared in interface AnnotatedElement

`getAnnotationsByType`, `getDeclaredAnnotation`, `getDeclaredAnnotationsByType`, `isAnnotationPresent`

Method Details

getAnnotatedOwnerType

`default AnnotatedType getAnnotatedOwnerType()`

Returns the potentially annotated type that this type is a member of, if this type represents a nested type. For example, if this type is `@TA O<T>.I<S>`, return a representation of `@TA O<T>`.

Returns `null` if this `AnnotatedType` represents a top-level class or interface, or a local or anonymous class, or a primitive type, or void.

Returns `null` if this `AnnotatedType` is an instance of `AnnotatedArrayType`, `AnnotatedTypeVariable`, or `AnnotatedWildcardType`.

Implementation Requirements:

This default implementation returns `null` and performs no other action.

Returns:

an `AnnotatedType` object representing the potentially annotated type that this type is a member of, or `null`

Throws:

`TypeNotPresentException` - if the owner type refers to a non-existent class or interface declaration

`MalformedParameterizedTypeException` - if the owner type refers to a parameterized type that cannot be instantiated for any reason

Since:

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getType

`Type getType()`

Returns the underlying type that this annotated type represents.

Returns:

the type this annotated type represents

getAnnotation

`<T extends Annotation> T getAnnotation(Class<T> annotationClass)`

Returns this element's annotation for the specified type if such an annotation is *present*, else null.

Note that any annotation returned by this method is a type annotation.

Specified by:

`getAnnotation` in interface `AnnotatedElement`

Type Parameters:

`T` - the type of the annotation to query for and return if present

Parameters:

`annotationClass` - the Class object corresponding to the annotation type

Returns:

this element's annotation for the specified annotation type if present on this element, else null

Throws:

`NullPointerException` - if the given annotation class is null

getAnnotations

`Annotation[] getAnnotations()`

Returns annotations that are *present* on this element. If there are no annotations *present* on this element, the return value is an array of length 0.

The caller of this method is free to modify the returned array; it will have no effect on the arrays returned to other callers.

Note that any annotations returned by this method are type annotations.

Specified by:

`getAnnotations` in interface `AnnotatedElement`

Returns:

annotations present on this element

getDeclaredAnnotations

`Annotation[] getDeclaredAnnotations()`

Returns annotations that are *directly present* on this element. This method ignores inherited annotations. If there are no annotations *directly present* on this element, the return value is an array of length 0. The caller of this method is free to modify the returned array; it will have no effect on the arrays returned to other callers.

Note that any annotations returned by this method are type annotations.

Specified by:

`getDeclaredAnnotations` in interface `AnnotatedElement`

Returns:

annotations directly present on this element

Report a bug or suggest an enhancement
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